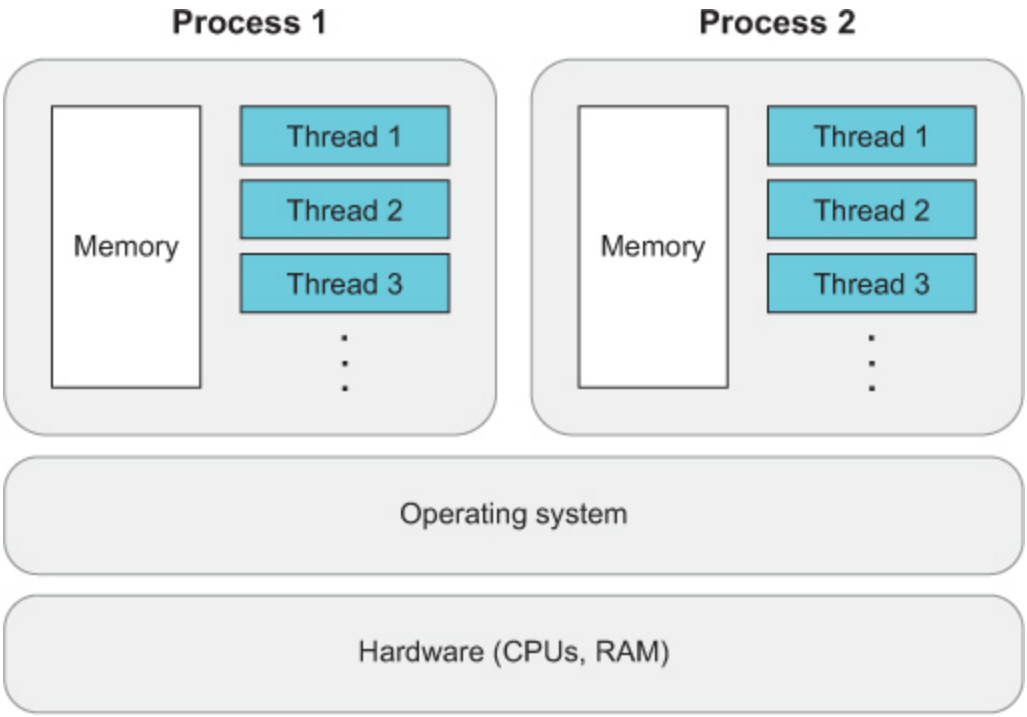


Processes are an abstraction of the underlying CPU hardware created by the operating system. If you have two CPUs, you can run two simultaneous processes. However, the operating system will let you spawn more than two virtual processes, and it will figure out how to multitask between them. Each process has its own memory address space and can have multiple threads (tasks). While one thread is waiting for an external process to finish (such as an input/output operation), the OS can let another thread run. This maximizes the use of whatever CPUs you have.

Multithreading is like when people multitask: they can work on only one thing at a time but they switch between different tasks while another task is idle. Therefore, tasks are not truly performed simultaneously with multithreading; it is a software-level mechanism to improve efficiency in running multiple computations. Multithreading is really effective when your task requires a lot of input/output operations, such as reading and writing data to the hard disk. When data is being read into RAM from the hard disk, computation on the CPU is idle as it waits for the required data, and the operating system can use that idle CPU time to work on a different task and then switch back when the I/O operation is done.